Quantification: 535_z13c1-2
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PRE-VALIDATION: no number here has been checked against manual (ImageJ) counts.
PIXEL UNITS: lengths px, areas px^2, densities px^-1. The images carry no micron calibration.
Statistics are over interior lacunae only (those not touching the frame edge).

Per image
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measure	value
lacuna count	23
interior lacuna count	20
frame-edge lacuna count	3

Interior lacunae only: mean, median and sample SD (n = 20)
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measure	kind	mean	median	sd	units
area_px2	lacuna shape	2539.15	1363.5	2593.51	px^2
major_axis_length_px	lacuna shape	81.7348	77.5147	46.6748	px
minor_axis_length_px	lacuna shape	37.3237	30.439	19.9187	px
aspect_ratio	lacuna shape	2.1914	2.2373	0.667	unitless
eccentricity	lacuna shape	0.8456	0.8944	0.1138	unitless
solidity	lacuna shape	0.7904	0.7775	0.0835	unitless
roots_count	headline	7.8	6.5	5.4541	count
ring_length_r30_px	headline	237.25	195	164.221	px
ring_length_r60_px	reported	472.15	390	310.151	px
owned_length_px	ownership-dependent	561.752	361.164	507.824	px
edge_count	ownership-dependent	27.65	16.5	26.898	count
mean_edge_length_px	ownership-dependent	22.4486	21.0474	9.1829	px
perimeter_px	normalised	257.918	205.258	182.397	px
ring_area_r30_px2	normalised	8353.1	7349	4493.18	px^2
ring_area_r60_px2	normalised	17278.9	17988.5	8436.35	px^2
in_frame_fraction_r30	normalised	1	1	0	unitless
in_frame_fraction_r60	normalised	0.9977	1	0.008	unitless
ring_density_r30	normalised	0.0267732	0.0283718	0.00860035	px^-1
ring_density_r60	normalised	0.0261594	0.0245539	0.00786176	px^-1
roots_per_100px_perimeter	normalised	3.1259	2.8767	1.5547	per 100 px
ring_attached_length_r30_px	network v2	205.6	171.5	156.188	px
ring_attached_length_r60_px	network v2	360.6	242.5	304.712	px
ring_length_w_r30_px	network v2	267.268	219.803	187.018	px
ring_length_w_r60_px	network v2	533.236	436.369	352.57	px
sholl_crossings_r10	network v2	6	5	4.5073	count
sholl_crossings_r20	network v2	6	6	3.5689	count
sholl_crossings_r30	network v2	6.8	6	4.3601	count
loop_count	loops, report only	1.25	1	1.6504	count
loop_edge_share	loops, report only	0.1166	0.1029	0.1213	share
ring_width_mean_r30_px	width	7.2325	7.5184	1.303	px

Per field
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measure	value	units
analysed_area_px2	988802	px^2
total_skeleton_length_px	20805	px
canalicular_length_density_per_px	0.0210406	px^-1
junction_count	381	count
junction_density_per_px2	0.00038531	px^-2
lacuna_count	23	count

lacunae_per_px2	2.326e-05	px^-2
skeleton_component_count	339	count
mean_component_length_px	61.3717	px
median_component_length_px	20	px
field_length_density_w_per_px	0.023686	px^-1
field_density_without_flagged_per_px	0.0205717	px^-1
field_density_in_roi_per_px		px^-1
unattached_length_px	799	px
unattached_component_count	48	count
canalicular_length_density_attached_per_px	0.0210406	px^-1
field_length_density_with_unattached_per_px	0.0218487	px^-1
loop_count	29	count
loop_edge_share	0.0852	share
edge_count	1103	count

Canalicular width, the new measure

measure	value	units
width_mean_px	7.3607	px
width_median_px	8	px
width_p10_px	4	px
width_p90_px	10.7703	px
width_pixels_used	19466	count

Canalicular width: 2 x the Euclidean distance transform of the canalicular mask at each skeleton pixel. Measured only on skeleton pixels on the canalicular mask: the lacuna buffer and the dilated vascular mask are excluded. The threads are about 2 to 6 px wide, so the width is close to the pixel grid and quantised in steps of about 1 px, it moves with the threshold (a lower cut widens every thread), and it has not been validated against any manual measurement.

Lacuna shape

lacuna_id	area_px2	major_axis_length_px	minor_axis_length_px	aspect_ratio	eccentricity	solidity	orientation_rad	centroid_row_px	centroid_col_px	on_border
1	520	50.9524	21.26	2.3966	0.9088	0.6061	0.134	15.2558	959.056	True
2	509	42.4483	19.3806	2.1902	0.8897	0.7313	1.0605	62.053	174.511	False
3	3449	94.2445	51.4301	1.8325	0.838	0.8008	-1.3442	87.5868	746.098	False
4	434	32.4903	18.5735	1.7493	0.8205	0.7993	1.0377	151.09	693.419	False
5	1518	84.7212	28.0385	3.0216	0.9436	0.7609	0.4889	372.306	481.817	False
6	916	42.5998	32.8395	1.2972	0.637	0.7829	0.4554	434.044	377.415	False
7	6729	150.978	65.1171	2.3186	0.9022	0.7721	-0.3046	496.001	291.95	False
8	3320	105.162	44.1209	2.3835	0.9077	0.7721	-1.4464	464.635	577.918	False
9	3645	102.737	49.6538	2.0691	0.8754	0.8697	0.3597	580.157	513.438	False
10	2801	98.9222	38.3287	2.5809	0.9219	0.8173	0.524	580.027	151.776	False
11	9054	174.453	85.8111	2.033	0.8707	0.6586	0.1605	649.475	679.118	False
12	1209	70.3083	25.9548	2.7089	0.9294	0.8103	-1	737.055	638.021	False
13	3256	105.273	45.222	2.3279	0.903	0.7485	-1.2191	649.232	325.526	False
14	524	31.117	22.6042	1.3766	0.6872	0.8748	-1.3629	760.422	888.611	False
15	597	63.166	16.0382	3.9385	0.9672	0.6958	-1.4137	764.35	853.412	False
16	543	31.4075	23.6295	1.3292	0.6588	0.9282	0.087	780.65	298.357	False
17	455	27.4749	22.5328	1.2193	0.5722	0.9362	1.1384	784.457	318.629	False
18	7705	171.941	75.2669	2.2844	0.8991	0.7155	-1.3369	837.132	645.403	False
19	2929	114.614	39.6482	2.8908	0.9383	0.6816	1.0561	854.515	315.604	False
20	580	52.7819	21.5044	2.4545	0.9132	0.716	1.2958	848.429	862.598	False
21	610	37.8566	20.7792	1.8218	0.8359	0.937	-0.7458	884.313	871.107	False
22	2259	90.1772	47.5611	1.896	0.8496	0.6829	0.6436	985.683	487.498	True
23	6212	244.279	36.0383	6.7783	0.9891	0.7991	-1.5538	1007.31	703.743	True

Network per lacuna

lacuna_id	roots_count	ring_length_r30_px	ring_length_r60_px	owned_length_px	edge_count	mean_edge_length_px	ring_width_mean_r30_px
1	7	169	259	182.439	10	18.2439	5.8346
2	6	186	464	511.245	24	21.3019	7.0114
3	9	380	909	1423.63	74	19.2382	7.2345
4	4	140	265	330.492	15	22.0328	5.721
5	7	181	367	123.953	11	11.2685	5.2487
6	11	234	413	769.997	33	23.3332	7.4626
7	17	485	939	1529.41	84	18.2072	8.2352
8	7	261	359	225.024	12	18.752	6.6633
9	3	204	417	202.64	7	28.9485	8.5039
10	12	263	534	499.031	24	20.793	7.9127
11	13	518	1025	1538.8	68	22.6294	7.8684
12	5	72	91	211.409	4	52.8523	5.8056
13	13	390	559	403.948	18	22.4416	7.669
14	1	95	292	194.196	8	24.2745	7.5743
15	4	72	225	175.569	11	15.9608	6.8691
16	2	86	206	123.669	7	17.667	8.7709
17	3	75	138	391.836	20	19.5918	8.2757
18	17	514	1013	1167.06	74	15.7711	7.6634
19	17	461	858	1130.38	51	22.1643	6.8189
20	4	73	172	269.51	7	38.5014	9.4276
21	1	55	197	13.2426	1	13.2426	3.9128
22	6	270	480	255.208	17	15.0122	8.2074
23	8	288	673	548.86	31	17.7052	8.3581

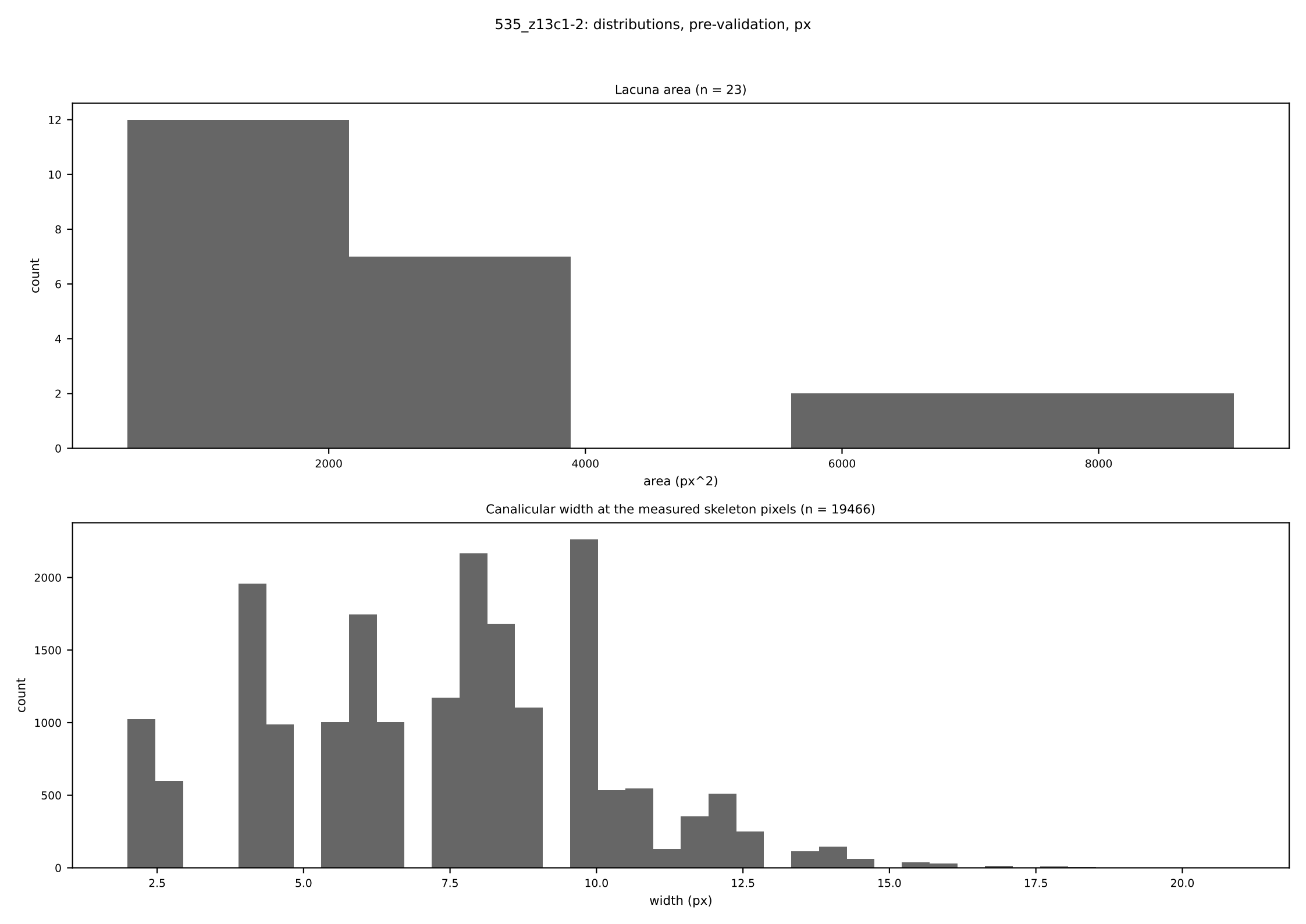
Normalised per lacuna

lacuna_id	perimeter_px	ring_area_r30_px2	ring_area_r60_px2	in_frame_fraction_r30	in_frame_fraction_r60	ring_density_r30	ring_density_r60	roots_per_100px_perimeter
1	153.297	4649	11757	0.6846	0.6173	0.0363519	0.0220294	4.5663
2	125.882	6125	17204	1	0.9656	0.0303674	0.0269705	4.7664
3	311.848	10440	22859	1	0.9885	0.0363985	0.0397655	2.886
4	104.225	4890	11875	1	1	0.0286299	0.0223158	3.8378
5	215.622	8573	18773	1	1	0.0211128	0.0195494	3.2464
6	145.782	5773	12342	1	1	0.0405335	0.033463	7.5455
7	433.103	13694	24741	1	1	0.035417	0.0379532	3.9252
8	326.563	11088	20860	1	1	0.023539	0.01721	2.1435
9	270.936	10229	21506	1	1	0.0199433	0.0193899	1.1073
10	264.007	10205	23754	1	1	0.0257717	0.0224804	4.5453
11	731.429	16860	32093	1	1	0.0307236	0.0319384	1.7773
12	194.894	5275	8641	1	1	0.0136493	0.0105312	2.5655
13	328.877	11820	23768	1	1	0.0329949	0.023519	3.9528
14	99.6335	3693	9318	1	1	0.0257243	0.0313372	1.0037
15	147.397	4813	9956	1	1	0.0149595	0.0225994	2.7138
16	96.0833	3059	6126	1	1	0.0281138	0.0336272	2.0815
17	85.734	2578	5393	1	1	0.0290923	0.0255887	3.4992
18	655.553	17714	33378	1	1	0.0290166	0.0303493	2.5932
19	380.475	11026	23669	1	1	0.0418103	0.03625	4.4681
20	139.498	4945	8638	1	1	0.0147624	0.019912	2.8674
21	100.811	4262	10684	1	1	0.0129047	0.0184388	0.992
22	275.764	8026	17251	0.7864	0.7193	0.0336407	0.0278245	2.1758
23	597.973	11032	21934	0.5965	0.5686	0.0261059	0.030683	1.3379

Network v2 per lacuna

lacuna_id	ring_attached_length_r30_px	ring_attached_length_r60_px	ring_length_w_r30_px	ring_length_w_r60_px	sholl_crossings_r10	sholl_crossings_r20	sholl_crossings_r30

1	145	160	197.238	299.806	3	4	2
2	157	275	213.51	530.659	3	4	7
3	323	729	426.848	1036.96	10	9	12
4	117	210	150.426	294.551	3	6	5
5	99	121	202.894	411.789	4	6	7
6	224	371	262.066	470.789	7	6	6
7	452	894	551.931	1065.34	13	10	14
8	189	210	288.309	393.492	6	12	6
9	186	318	226.095	460.948	7	8	5
10	249	430	303.078	603.115	7	8	8
11	499	933	585.902	1161.42	14	12	14
12	68	87	83.2548	106.397	1	2	1
13	316	361	447.475	633.428	8	8	11
14	83	194	111.054	333.492	3	3	4
15	72	136	77.8701	245.368	2	1	1
16	63	118	94.1838	238.338	2	1	5
17	47	72	80.2843	148.083	1	3	2
18	487	866	576.718	1141.13	15	9	13
19	394	766	524.961	973.051	10	8	11
20	73	107	80.6985	193.125	4	2	2
21	14	14	57.799	223.238	0	2	2
22	254	354	304.522	540.174	7	9	7
23	243	422	315.865	751.027	7	10	8


535_z13c1-2: distributions, pre-validation, px
Lacuna area (n = 23)
count
area (px^2)
0
2
4
6
8
10
12
2000
4000
6000
8000
Canalicular width at the measured skeleton pixels (n = 19466)
count
width (px)
0
500
1000
1500
2000
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0